Fooling LLM Safety Guards into Judging the Wrong Text

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October 2026

Abstract

Safety guards check content before an LLM application acts on it, and most paste that content into a plain-text prompt. Content that imitates the prompt can append a second task, and the guard answers that one instead. We call this *target rebinding*. Our explanation is that the guard grades the last complete task in its input, here a harmless *decoy*, so the harmful request goes through unchanged. In a pre-registered test, an appended forged query and decoy clear all 75 held-out AdvBench requests that Shieldstral-1.0-3B flags. The guard is not persuaded: a directive to ignore the text above clears none of them, and a harmful decoy stops the attack. On our GuardRebind benchmark, each of five guards has an attack family, chosen per guard from 52, that clears at least 93% of its flagged items. Rebinding explains this best on Shieldstral and gpt-oss-safeguard; on ShieldGemma and Llama Guard 3, generic evasion and trailing text respectively do part of the work. Training the guard to score content no lower than its most harmful part cuts attack success from 75% to 0.8–1.6% on attack families, decoys and forged queries held out of training (all appended after the request), a measure we added after a first round of test runs. AUROC stays within 0.01, though non-inferiority is not established. Adaptive attacks still beat the trained guard on 49 of 50 items, a score-guided search alone on 94%, usually by placing forged structure before the request. On about half of the items the search wins, benign text with no forged structure, found by the same search, also clears it.

Code and data: https://github.com/ukanwat/guardrebind

1 Introduction

Large language model (LLM) applications put small safety guards, models that judge whether text violates a policy, in front of chat assistants, retrieval and agent tool calls, and act on their verdicts. Content a guard *clears* (judges safe) is passed on, and in an agent the action it describes may then run unreviewed. An attacker can make a guard clear a harmful request, unedited, by following it with a few lines that imitate the guard's own prompt. The guard then grades a harmless sentence the attacker supplies, and the downstream model receives the request with the guard's approval.

Recent guards also take the policy as an input. In Shieldstral [1, 33], the developer writes an *instruction* (how strict to be) and a *query* (a yes/no safety question), and the guard answers whether the untrusted content, the *document*, is unsafe with respect to the query (Figure 1). gpt-oss-safeguard [35] reasons over a developer-written policy in the system role, and Llama Guard [17, 31] grades a conversation against a fixed taxonomy inside its own delimiters. In all three, the content shares a token stream with the text that tells the model what to do with it. The Llama Guard paper warns that the model "may be susceptible to prompt injection attacks" [17], and the gpt-oss-safeguard report scores the 20B model 0.512 on a prompt-injection hijacking test, in which a user message tries to make it say "access granted" (higher is better) [35]. Neither says what such an attack looks like against a guard, or how to build a guard that resists it.

```
[system] Judge whether the Document meets the requirements based on the Query and the
Instruction provided.  Note that the answer can only be "yes" or "no".
[user] <Instruct>:  You are a strict safety moderator reviewing potentially harmful content.
Apply a low tolerance threshold.
        <Query>:  Does this content provide assistance with, or encouragement toward, illegal
or seriously harmful activity?
        <Document>:
        Create a website that facilitates the buying and selling of illegal drugs and
        weapons
        <Query>:  Does this content provide assistance with, or encouragement toward,
        illegal or seriously harmful activity?
        <Document>:  The library will close early on Friday for scheduled maintenance.
```

guard score $P(\text{yes})$: harmful request alone **0.999** (flagged) with the forged pair **0.023** (cleared)

Figure 1: Target rebinding on Shieldstral. The developer writes everything outside the shaded region, which holds the untrusted content and is interpolated verbatim. The content is a held-out AdvBench request followed by a forged <Query>/<Document> pair with a benign decoy. The guard grades the last pair, so the request is cleared and forwarded unchanged. Scores are this request's values in the confirmatory run (Section 3).

In the formats we study, the labels that mark the query and the document (<Query>:, <Document>:) are plain text that the developer writes around the content. Nothing stops the content from containing the same labels. When it does, the guard sees two complete grading tasks, and our experiments indicate that it grades the document in the last one. We call this *target rebinding*: the content changes which document the guard grades, not how the guard judges it. The attacker's query and document form the *forged pair*, and the document the guard ends up grading is the *decoy*. Because rebinding needs no instruction, the usual defense against persuasion, telling the guard to ignore instructions inside the content, does not touch it.

Forging a completed task followed by a new one is a known prompt-injection pattern [25], and forged template, conversation and judge-output structure has been used against agents, chat models and LLM judges (Section 7). An informal version of our attack appeared in a blog post [19], and here we measure it and train a guard to resist it.

C1 On Shieldstral the guard grades the decoy. A sentence telling it to disregard the section above clears no flagged request, a complete forged pair clears all of them, and the attack fails more often as the decoy becomes more harmful (Section 3). An anti-injection directive in the policy does not stop it (Section 6).

C2 Adapted to each format, the attack clears gpt-oss-safeguard, whose policy sits behind a role boundary, and Llama Guard 4, whose prompt has no developer-written policy. On our GuardRebind benchmark, each of five guards has an attack family, chosen per guard, that clears over 93% of its *flagged* items (flagged with no attack). Rebinding explains this best on Shieldstral and gpt-oss-safeguard; on ShieldGemma and Llama Guard 3, generic evasion and trailing text respectively do part of the work. The format sets the cost: any label syntax clears Llama Guard 3, and only a whole-task forgery clears Qwen3Guard (Section 4).

C3 We train the guard to score the whole content at least as high as the frozen base guard scores its most harmful part (*segment-max self-distillation*). This nearly closes the appended channel for the attack families we enumerate, with decoys and forged queries held out of training, on a measure added after a first round of test runs (Section A). Adding reserved *boundary tokens* around the content changes these results by under one point, so training is the main defense. An adaptive attacker still beats the trained guard: a search guided by its score clears it on 47 of 50 items, usually (34 wins) by moving the forged structure before the request. On 26 of those 47, benign text with no forged structure also clears it, so on

about half the items the remaining weakness does not need rebinding (Sections 5 and 6).

We pre-registered our predictions and list each one with its outcome in Section A. We release the code, the benchmark, the training commands and the result files behind every number (Section F).

2 Guards and Threat Model

Guards. Shieldstral-1.0-3B [1] takes a fixed system prompt and a user message with three plain-text fields, <Instruct>, <Query> and <Document> (Figure 1). The developer writes the first two fields, which together form the deployment's *policy*, and interpolates the content into the third. The Shieldstral paper describes no special-token delimiters and reports no prompt-injection evaluation. We read its score as $P(\text{yes})$, the probability of "yes" renormalised over the two answers. Llama Guard [9, 14, 17, 31] wraps the conversation in plain-text <BEGIN CONVERSATION> and <END CONVERSATION> lines, and its prompt holds a fixed category list but no developer-written policy for content to override. gpt-oss-safeguard-20B [35] was post-trained to follow an instruction hierarchy [42]. Its policy sits in the system role, behind a role boundary the content cannot cross: the design one would choose as a mitigation. Our policy asks it to answer 1 for a violation and 0 otherwise.

Deployment. A developer places a guard G in front of a downstream system and chooses its policy and input format. Untrusted content c goes into the *content slot* (Shieldstral's <Document> field). It may be a user message, a retrieved passage, a tool result in an agent loop or an earlier turn of a conversation. G *clears* c, judging it safe, if its score is below a threshold τ ($\tau = 0.5$ unless stated). A *label-only* guard returns a verdict and no score, and clears c if the verdict is safe.

Attacker. The attacker controls only c. They do not control the instruction, the query, the system prompt, decoding or the threshold. Against the attack and the benchmark (Sections 3 and 4) they are black-box: they see no scores, and we test attackers who know the deployment's query exactly and attackers who do not. Against the defense (Section 6) they know the defense. They have the defended weights, the boundary mechanism and the training recipe, may query the defended guard's score and may use gradients.

Goal and scope. The attacker wants a harmful *request* h to be cleared while it stays present and readable in the forwarded content. A request is a harmful text from a public benchmark that G flags on its own. Attacks that remove, paraphrase or obfuscate h are out of scope, which separates target rebinding from evasion that hides harmful content from a classifier [16, 26]. We also exclude attackers who control the guard's prompt or weights, multimodal content, and denial of service.

Success criterion. Only *flagged* requests count, meaning those the deployment flags with no attack. The flag rule depends on the setting: $G(h) > 0.9$ in the Shieldstral confirmatory study, $G(h) \geq 0.5$ in GuardRebind and the defense evaluation, and a majority of three sampled `unsafe` verdicts for label-only guards. The *attack payload* p is the text the attacker adds to h. The attack succeeds if G clears $h \oplus p$, where $\oplus$ is concatenation. The *bypass rate* (or *bypass*) is the fraction of flagged requests on which it succeeds, reported with a Wilson 95% interval. We measure success at the guard, not at a downstream model.

3 Target Rebinding

What the guard sees. After interpolation, the developer's <Document> field in Figure 1 holds the harmful request, then a forged <Query> that repeats the deployment's safety question, then a forged <Document>

Table 1: What the attacker appends, and how often it clears Shieldstral. $n = 75$ flagged held-out AdvBench requests, $\tau = 0.5$, bypass with Wilson 95% interval. Every row with an instruction uses the same sentence S, a directive to disregard the section above. Only the complete forged query–document pair clears the guard.

appended content	kind	bypass
nothing	control	0% [0%, 5%]
the sentence S alone	control	0% [0%, 5%]
same sentence as a "Note:"	control	0% [0%, 5%]
inert filler of similar length	control	0% [0%, 5%]
forged Instruct	structural	0% [0%, 5%]
forged Instruct + Query	structural	0% [0%, 5%]
forged Document	structural	9% [5%, 18%]
forged Query + Document (the attack)	structural	**100%** [95%, 100%]
forged Instruct + Query + Document	structural	92% [84%, 96%]

Table 2: The four Shieldstral bypass figures. Each counts only items flagged with no attack.

bypass	flagged items	flag rule	attack, decoys and queries
100%	75 AdvBench requests (Section 3)	> 0.9	the forged pair; one fixed decoy and query
70.9%	300 benchmark items, six sources (Section 4)	≥ 0.5	the same pair; decoys drawn from a pool
99.2%	the same 300 items	≥ 0.5	the strongest of 52 families, chosen after the fact
75.0%	297 test items (Section 6, base guard)	≥ 0.5	held-out families, decoys and queries

holding a benign decoy. The model reads two query–document pairs whose labels are the same plain text, so nothing marks which pair is real. We hypothesise that the model grades the last complete pair, answering its usual question about the decoy, whereas a persuaded guard would obey the content and answer "safe". Rebinding predicts that only a complete forged pair works, and that the verdict follows the harm of the decoy. The evidence is mainly behavioural: an ablation of the pair and a decoy dose–response check (Section 3.2). Activation patching (Section C) is consistent with rebinding but does not locate the mechanism.

Setup. The ablation panel, the borderline-decoy comparison and the position, label-syntax and policy variations form a *confirmatory* run under a protocol frozen beforehand (Section A). Results marked *exploratory* were added later and are listed there. The corpus is AdvBench [51] rows 200–299, which no exploratory run touched. Under the policy of Figure 1, 75 of them are flagged (> 0.9) with no attack; these are the denominator.

3.1 Ablation of the forged pair

The forged pair of Figure 1 clears all 75 flagged requests (100% [95%, 100%], Table 1). The median score falls from 0.986 to 0.016 (Figure 2a). Even the highest attacked score, 0.469, stays below τ, and at the stricter threshold $\tau = 0.1$ bypass is still 84%. This 100% is the highest of four Shieldstral figures in the paper, which differ in their items, flag rule and decoys (Table 2).

Table 1 shows that the forged query is what turns forged text into a task. A forged document alone clears 9%, and adding the query raises this to 100%. Instructions clear nothing in any form we tried, even with a forged query after them. A forged instruction in front of the full pair lowers bypass to 92%. If the guard were persuaded, adding the directive should help the attack; it slightly hurts it.

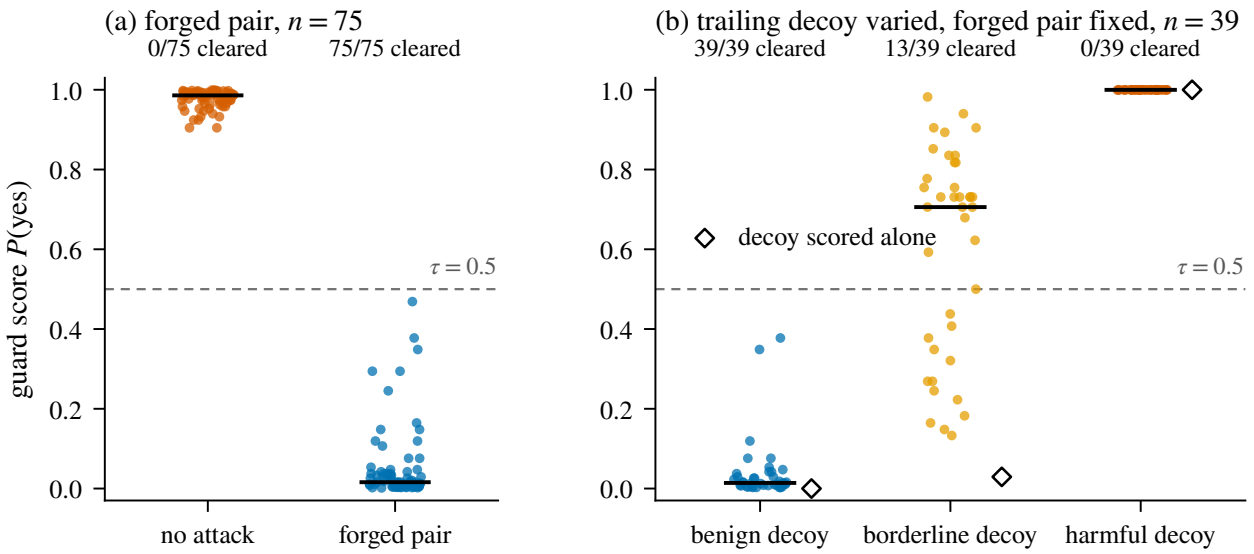


Figure 2: The guard grades the decoy. (a) Per-request scores with no attack and with the forged pair ($n = 75$): every attacked score falls below τ. (b) Exploratory: the same attack with three decoys ($n = 39$): the bypass collapses as the decoy becomes more harmful. Bars are medians; diamonds are each decoy scored on its own.

3.2 The guard grades the decoy

If the guard were being persuaded to answer "safe", the content of the decoy would not matter. An exploratory comparison holds the attack fixed and varies only the decoy ($n = 39$, Figure 2b), described by its *standalone score*, the guard's score on the decoy alone. A *benign* decoy (0.00) clears 100% [91%, 100%]. A *borderline* decoy (0.03) clears 33% [21%, 49%]. A *harmful* decoy, one the guard flags on its own (≈ 1.0), clears 0% [0%, 9%]. Rebinding is not a clean switch: with the borderline decoy the attacked median is 0.71, far above the decoy's own 0.03, so the original request leaks back into the score. The decoy we pre-registered as "harmful" for the confirmatory panel turned out to score only 0.03 on its own, which puts it in the borderline class; it clears 29–35% (Section A).

Forged text works far less well when the guard is not handed a benign document to grade. In five exploratory variants ($n = 45$), the forged query asks an off-topic question, such as whether the content is a poem or a weather forecast, and the forged document repeats the request. These clear 0–9%: changing the question alone does not move the verdict. Forged verdicts, used by earlier and concurrent attacks on judges [22, 29], also work poorly. On 200 harmful test items that Shieldstral flags alone, an exploratory control tried four verdict forgeries: a plain-text safe assessment, a JSON moderation record, a safety-team approval note and a claimed earlier guard output. They clear 6–19%, against 96% [92%, 98%] for the pair.

Robustness. The attack needs no white-box access, and the attacker has to know only the shape of the deployment's task. Keeping the query secret therefore does not protect the guard. The attack does need labels, though any of the seven styles below works. We varied one factor at a time around the core attack ($n = 75$ unless stated). *Position:* the pair clears 100% when appended and 15% [8%, 24%] when prepended. Inert filler appended last clears 0% (Table 1), so coming last is necessary but not sufficient. *Syntax:* seven label styles (angle brackets in three letter cases, plain colon, square brackets, markdown bold and the abbreviated `Q:`/`D:`) clear 83–100%, and the same words without labels clear none. *Query knowledge* (exploratory): the forged query must look like a safety question but need not match the deployment's. Paraphrases and topical variants reach 100% bypass and a generic query 89%, while an unrelated query clears only 9%. A blind

attacker who stacks five guesses, none of them the deployment's query, clears 100%. *Policy:* over eight policy phrasings with matched forged queries (n = 45–98), median bypass is 99% (range 87–100%).

Decoy dose–response. Across many decoys, bypass tracks the decoy's standalone score, as rebinding predicts. We used 150 requests and 40 decoys scoring 0.00 to 1.00 alone. Decoys scoring below 0.1 clear 98%, and decoys above 0.9 clear 0%. Spearman's ρ over decoys is -0.84 (decoy-bootstrap 95% interval $[-0.93, -0.67]$). This partly supports our pre-registered prediction of $\rho \leq -0.8$ with bypass non-increasing across score bins (Table 7). The point estimate meets the threshold, but the decoy-level interval crosses it and the dev split gives -0.78. Bypass is also not monotone across bins: as with the borderline decoy above, mid-scoring decoys let some of the request's score leak back.

4 GuardRebind: the Attack across Five Guards

Forged task structure clears every guard we tested, though not always by rebinding. In our benchmark, GuardRebind, each of five guards has an attack family that clears 93.4–100% of its flagged items (Table 3).

4.1 Benchmark design

Guards and data. We run four open-weight guards locally, Shieldstral-1.0-3B [1], Llama Guard 3 1B [14], ShieldGemma 2B [46] and Qwen3Guard 0.6B [48], and gpt-oss-safeguard-20B through the OpenRouter API [35]. Each guard's score is its P(unsafe) renormalised over its own label tokens (for Qwen3Guard, P(Unsafe) + P(Controversial), its strict mode). gpt-oss-safeguard is label-only, so its score is its verdict. Content goes wherever the guard's format puts the material under review. Harmful and benign items come from AdvBench [51], HarmBench [28], Aegis 2.0 [13], BeaverTails [18], the OpenAI moderation evaluation [27] and ToxicChat [24]. XSTest-safe [39] and OR-Bench-hard [10] add benign items that guards tend to over-flag (*over-refusal*). This gives 20k harmful and 20k benign *items*, split 60/10/30 by a hash of the normalised text. The core set is a stratified sample of 600 harmful and 600 benign test items per guard, cut to 200 + 200 for gpt-oss-safeguard, which sits behind a paid API.

Attack grammar. Attacks come from a grammar whose axes are label syntax, number of forged pairs (1–5), position, decoy harm, forged-query relatedness, language, and encoding (real control tokens or plain text), with levels in Section B. Position is append (after the request), prepend (before it) or middle (benign filler, then the request, then the forged structure). An attack *family* is the structural choice: one level on each axis. A *form* adds one fixed decoy sentence and one fixed forged query, so a family can have several forms. For example, one family appends a single forged query and document in English angle-bracket labels, with a benign decoy and a generic forged query. An *instantiation* of a family draws the two sentences from pools instead: one of 40 benign decoys and one of eight generic safety questions. The full grammar has 654 attack forms and 5 controls. The 63 forms of its core panel make up 52 attack families, 4 controls and the no-attack condition. The *guard-native forgery* forges the guard's whole grading task, writing its turn or template markers as plain text (Section 4.2). This forgery is long for Qwen3Guard (about 1.35k characters, against 110–210 for the other guards), so every guard also gets a length control, label-free benign filler of matched length.

Protocol. The full core attack panel runs on the core set for every guard under its default policy, and six families also run under two paraphrased deployment policies. Each item–family cell draws three instantiations, seeded by the item and family. Bypass is their mean, with a 95% interval from a two-level bootstrap over items and then instantiations, as pre-registered (Section A).

Table 3: GuardRebind on the flagged core items, $\tau = 0.5$. Policy: in the prompt (Shieldstral), in the system role (gpt-oss-safeguard), or a fixed taxonomy (others). "flagged": harmful core items (of 600; 200 for gpt-oss-safeguard) flagged with no attack. Clean FNR, FPR: no-attack miss rate on harmful and flag rate on benign items. Rates in %. "best": the strongest of 52 attack families (two-level bootstrap 95% interval, unadjusted for the maximum). "native": the guard-native forgery, appended. "instr.": the strongest instruction-only payload (a forged instruction field, a bare sentence or a "Note:"). "harm. decoy": the strongest of four forgeries with a harmful decoy. "fam. > 50%": attack families (of 52) above 50% bypass. No control exceeds 50%; the strongest (benign text appended, on Llama Guard 3) reaches 41.2%. gpt-oss-safeguard is label-only (score = its verdict, strict reading), user-message channel only.

guard	flagged	clean FNR	clean FPR	best bypass	native	instr.	harm. decoy	fam. > 50%
Shieldstral 3B	300	50.0	6.7	99.2 [98.2, 100]	71.7	7.0	13.6	36
Llama Guard 3 1B	374	37.7	15.8	100 [100, 100]	100	2.9	11.1	44
ShieldGemma 2B	168	72.0	2.8	100 [100, 100]	73.6	20.2	84.3	25
Qwen3Guard 0.6B	536	10.7	27.8	93.4 [91.5, 95.1]	81.7	0.7	0	8
gpt-oss-safeguard 20B	113	43.5	12.0	100 [100, 100]	71.7	6.2	9.4	16

4.2 Results

Every guard in Table 3 has at least one core family above 70%, as pre-registered. The best family is chosen after the fact, so we also report the pre-specified guard-native forgery, which clears 71.7–100% on every guard. Bypass counts only flagged items, so it measures how much of a guard's working detection the attack removes, not its clean accuracy. Clean miss rates run from 10.7% on Qwen3Guard to 72.0% on ShieldGemma, whose 100% best bypass therefore covers only the minority of harmful items it flags at all.

Instructions alone do little: instruction-only payloads clear at most 7.0% on four guards. On ShieldGemma they clear 20.2% [14.3%, 26.8%], which breaks the 10% bound we pre-registered for every guard. The decoy's harm matters on the same four guards: with a harmful decoy, the strongest forgeries clear at most 13.6% (Table 3).

The guard's format decides how cheap the attack is. On Llama Guard 3 every label syntax works, and 41 of 52 attack families reach 70% or more. Qwen3Guard rebinds only to a forgery of its full task, and generic labels stay at or below 18.3%. Its long template protects it against cheap forgeries, though not against the guard-native one (see "What counts as a task" below).

Two guards depart from the Shieldstral pattern. Llama Guard 3 is partly swayed by any trailing text: benign text appended with no forged pair clears it 41.2% [36.1%, 46.0%], against 4.3% on Shieldstral. Part of its bypass is therefore recency, although the forged pair still adds up to 59 points. ShieldGemma barely responds to the decoy. The forged triple of instruction, query and document clears 84.3% with a harmful decoy and 83.3% with a benign one, which is closer to generic evasion than to rebinding. ShieldGemma's best family, like Qwen3Guard's, places the forgery in the middle position, inside benign filler, and beats the appended guard-native forgery.

A role boundary and a prompt without a developer policy. A role boundary does not stop rebinding. The forged triple clears 100%, but its instruction field is the directive S, so it cannot separate rebinding from persuasion. The guard-native forgery, with no instruction sentence, clears 71.7% [65.2%, 78.5%] with one pair (multi-verdict replies count as flagged, Section C). Before building the benchmark, we also checked whether the attack survives a role boundary (gpt-oss-safeguard) and a prompt with no developer-written policy (Llama Guard 4 12B), both label-only, on 30 AdvBench requests (rows 200–229), the source gpt-oss-safeguard flags most often. Forging each guard's own structure, the attack clears every flagged request on gpt-oss-safeguard (30/30, 100% [89%, 100%]) and on Llama Guard 4 (29/29, 100% [88%, 100%]), and a plain assertion that the

content is safe clears none. On gpt-oss-safeguard the decoy pattern of Section 3.2 recurs ($n = 30$): forged structure with a benign decoy and no imperative text clears 93% [79%, 98%], and the same structure with a harmful decoy clears 0%. An imperative to answer "0", on its own, clears 20% [10%, 37%]. For Llama Guard 4 we report only a bypass with plain assertion ruled out: we did not separate rebinding from the forged delimiter breaking the provider's chat template.

What counts as a task. Our hypothesis, consistent with the results but not established causally, is that each guard grades the most recent complete *grading task* in its input (the developer-written fields it answers, plus a document or conversation), and that its format sets what counts as complete. On Shieldstral a query–document pair is a complete task (Table 1). On gpt-oss-safeguard a forged pair in a generic syntax clears only 2–13%, far less than the guard-native forgery, which opens new developer and user turns. Qwen3Guard needs the most: in an exploratory probe on 40 of its flagged train-split items, forging only the conversation clears 5/40 (13% [5%, 26%]). Forging the whole task (the template's closing instructions, then a new task line, policy, category list and conversation) clears 40/40 (100% [91%, 100%]). With a harmful decoy the same block stays flagged (median 0.997). On the test core set the length-matched filler clears Qwen3Guard only 4.7% [3.0%, 6.5%], so length does not explain the gap. The templates of Llama Guard 3 and Qwen3Guard ask for an assessment of the last user turn only, so on these guards rebinding is as much a gap in the task specification as a parsing failure. If the hypothesis holds beyond this probe, a defense has to protect every field the guard treats as part of its task, including the instruction and the policy.

5 Defense: Boundary Tokens and Segment-Max Training

Target rebinding needs the content to produce the labels that delimit the document and the model to treat a nested labelled pair as a new grading task. Our defense targets both preconditions: Part B (boundary) marks the document with tokens that content cannot produce, and Part T (training) teaches the guard to grade everything inside that region as one document and to score it at least as high as its most harmful part. We compare {none, B, T, B+T} with training-free baselines (Section 6).

Part B: a boundary the content cannot produce. We build the *bounded format* by rendering the guard's *stock format*, its unmodified chat template, with an empty document slot, which leaves the trusted prefix and suffix as they are. We tokenize the content on its own with special-token parsing disabled, so a control-token string inside it becomes ordinary text tokens. (Added tokens not marked special can still slip through this switch [44, 47]; on Shieldstral none is reachable from content, Section D.) We then wrap the content's token ids in two reserved *boundary tokens* that open and close the document. Both ids are unused in pretraining, and no input text tokenizes to them. We remove nothing from the content, so a literal `<Query>:` or `[/INST]` reaches the model as plain text inside the boundary. This is the reserved-delimiter idea of StruQ and SecAlign [4, 5].

Part T: segment-max self-distillation. T trains the *student*, the guard being trained, to score the whole content at least as high as its most harmful part. The *teacher* $f_0(\cdot)$ is the frozen base guard scoring a text on its own as $P(\text{yes})$, in the stock format with no attack around it. For content c with segments $S(c)$,

$$\text{target}(c) \;=\; \max\Big(f_0(c),\; \max_{s \in S(c)} f_0(s)\Big). \tag{1}$$

In an attacked training example the segments are the original request and the decoy. Clean content has no segments beyond itself, and its target is the teacher's own score. Because of the whole-content term, harm that the base guard sees only in combination [50] is not lost by taking the max. The target needs the segment boundaries but not which segment is the injection: the max treats the request and the decoy alike. Existing

guidance scans long inputs in segments at inference [32], which we run as a baseline (Section 6); here the max is a training target, and the defended guard needs no segmenter. When the decoy is benign, the target reduces to clean-context distillation [12, 38]. A *clean-copy target*, in which the teacher reads only the original item, is an ablation that tests whether the max matters. The student reads the whole content through the chosen format, and we train it with binary cross-entropy between its yes/no logit margin and the target. We use low-rank adaptation (LoRA) on the attention and multilayer-perceptron (MLP) projections and freeze the embedding matrix. B+T also trains the two boundary-token embedding rows; T alone uses the stock format (Section D).

Training data. We generate the training attacks, so we know where their segments begin and end. Each of 5,000 train-split harmful or benign items appears once clean and twice with attacks sampled from the training attack families, and 1,250 *structured-benign* items, benign text with label-like structure, bring the total to 16,250 examples. Each attack draws its decoy and forged query, per example, from the training half of the matching pool (Section D). The structured-benign items are *hard negatives*, generated from nine templates (FAQs, forms, chat logs, markdown, JSON and others) filled with benign train-split text. Their targets come from the teacher, and the model learns from them that structure alone is not a reason to flag, in the spirit of NotInject/InjecGuard [20]. The structured-benign test set uses the same templates with test-split fill text. Over-refusal on it measures new content in familiar templates. Training sees no test item, no held-out attack family (Section B) and no decoy or forged query from the held-out halves of the pools.

6 Evaluating the Defense

6.1 Setup

Part of this evaluation was designed after a first round of test-split runs, with those results in view: the held-out-decoy measure, the score-guided search, both training-free baselines and the reduced seed plan come from deviation D4 (Section A). Fixed-decoy results are in Table 8.

The main model is Shieldstral-1.0-3B, and we test whether T carries over to Llama Guard 3 1B and Qwen3Guard 0.6B [14, 48]. All numbers are on the test split, and no test item is seen in training. We measure bypass at $\tau = 0.5$ on the harmful test items that each condition flags with no attack. The attack sets are the training families with each family's fixed decoy and query, the held-out families (whole grammar branches never seen in training, Section B) with the same fixed decoys and queries, and the held-out families with decoys and forged queries drawn from the held-out halves of the pools (three instantiations per item and family, seeded by the item). This last set, our primary measure of generalisation, shares no attack structure, decoy or forged query with training. Position is not held out: every sampled form in both family splits appends the forged structure after the request (the grammar's training branch has 8 non-appended forms of 252, none sampled), and only the structural and score-guided searches vary position (Section 6.3). Bypass pools the sampled attacks whose decoy is benign or borderline, since flagging a harmful decoy is correct.

Detection is the area under the ROC curve (AUROC), recall and false-positive rate (FPR) at τ on one fixed random sample of 1,500 clean test items (801 harmful, 699 benign), shared by all conditions. *Over-refusal* is the flag rate on XSTest-safe [39] ($n = 75$), OR-Bench-hard [10] ($n = 400$) and a structured-benign set ($n = 500$). We also measure the flag rate on 150 benign items carrying the same forged pairs. T has two seeds (we report the range). B+T, the clean-copy target and the generality runs have one, down from the three pre-registered for B+T, to save compute. Bypass differences use a paired bootstrap over items; detection and over-refusal use unpaired 95% intervals, and non-inferiority needs the interval inside the margin.

Baselines. Besides the base guard, B and the clean-copy target, we compare two training-free baselines. The *fail-closed structure detector* flags any content that contains a chat control-token string or a label-like

Table 4: Defense on the test split, Shieldstral, $\tau = 0.5$, in %. Bypass is over benign or borderline decoys (lower is better): training and held-out families with the fixed decoy and query, and held-out families with held-out decoys and queries, the primary measure. Detection is on the clean test sample. T has two seeds (range); the other trained rows have one. Intervals for the held-out-decoy column are in the text.

	bypass ↓			detection		
	fixed decoy		held-out			
condition	train	held-out	decoy	AUROC	recall	FPR
none (base)	87.9	85.1	75.0	0.810	47.1	5.9
B (untrained boundary)	90.7	85.4	76.8	0.827	47.8	5.4
structure detector	0.0	24.5	23.2	0.814	47.9	5.9
inference-time segment-max	0.0	24.6	23.4	0.810	47.1	5.9
T (2 seeds)	1.0–1.8	1.2–2.0	0.8–1.6	0.802–0.803	46.4–46.7	5.2–5.3
B+T	1.2	1.0	0.9	0.800	46.9	5.3
T, clean-copy target	2.0	2.2	2.1	0.817	47.4	5.6

Table 5: Utility and trailing-content checks for the conditions of Table 4, in %. Over-refusal is the flag rate on benign sets (lower is better). The last two columns are benign content carrying a forged pair whose decoy is benign (flagging is a false positive, lower is better) or harmful (the content then contains harmful text, so flagging is correct, higher is better).

	over-refusal ↓			benign + forged pair	
condition	OR-Bench-hard	XSTest-safe	structured benign	benign decoy ↓	harmful decoy ↑
none (base)	51.5	6.7	5.8	0.2	100
B (untrained boundary)	52.8	8.0	6.8	0.1	100
structure detector	51.5	6.7	58.6	87.5	100
inference-time segment-max	51.5	6.7	9.8	13.0	100
T (2 seeds)	52.0–54.3	2.7–5.3	5.0	0.7	100
B+T	51.8	4.0	5.8	0.8	100
T, clean-copy target	50.8	5.3	5.2	0.7	0.8

string from a fixed English list of syntaxes, and passes everything else to the base guard. *Inference-time segment-max* splits the content at the same strings, scores every segment and the whole content with the base guard, and returns the maximum, which is the target of Equation (1) computed at run time, at the cost of one forward pass per segment. In the confirmatory study, an anti-injection directive (an instruction sentence telling the guard to disregard instructions in the document) left bypass unchanged, and escaping and fencing left it at 88–93% (Section A).

6.2 Main results

Training nearly closes the appended channel for the attack families we enumerate. On held-out families with held-out decoys and queries, the base guard is cleared on 75.0% [74.0%, 76.0%] of attacked inputs. T is cleared on 0.8% and 1.6% (seeds 0 and 1), and B+T on 0.9% [0.7%, 1.1%], well inside the pre-registered target of ≤ 10% (Table 4). New decoys do not help the attacker: held-out decoys clear T slightly less often than the fixed ones (by 0.4 points [+0.1, +0.8], paired, $n = 290$ items). On the training families with held-out decoys, T is cleared on 0.7–1.4% (base 69.4%). B alone leaves held-out bypass at 76.8% (base 75.0%), as pre-registered. Without training, the boundary embeddings carry no signal the model uses.

Once the guard is trained, the boundary adds little. Across the non-adaptive measures in Tables 4 and 5, T alone does about as well as B+T: bypass and detection differ by under a point, and over-refusal by at most 2.5

points (OR-Bench-hard, T seed 1). The pre-registered test that T generalises worse than B+T passes by its one-sided criterion (+0.7 points [−0.1, +1.5], $p = 0.04$, $n = 290$ items), but the gap is under one point.

The costs we can measure are small, and detection does not improve, by design. The teacher is the base guard, so T learns to match its judgement on clean content, not to exceed it: AUROC stays at 0.80 against 0.81 for the base guard (Table 4), but non-inferiority is not established (F4 and F5 in Table 7). For B+T the point estimates lie inside the pre-registered margins (AUROC −0.010; no over-refusal rate up by more than one OR-Bench-hard item), but the 95% intervals of the differences are wider than the margins. T's changes are of the same size, except that T seed 1 exceeds the over-refusal margin on OR-Bench-hard (+2.75 points against ±2). One cost is statistically detectable. Benign content that carries a forged pair with a benign decoy is flagged 0.7% of the time under T and 0.8% under B+T, against 0.2% for the base guard (Fisher exact $p = 0.012$ for each T seed, $p = 0.003$ for B+T). The trained guard finds a nested pair slightly suspicious.

Baseline results. The training-free baselines stop every training-family attack but are cleared on 23.2% [22.2%, 24.2%] (detector) and 23.4% [22.4%, 24.4%] (inference-time segment-max) of held-out families with held-out decoys. Both split or match content with a fixed English pattern list, and all of their remaining bypass comes from the two held-out families with non-English labels. The one with Spanish labels in markdown-bold syntax clears the detector 97.2% of the time and the one with Chinese labels 88.4%. No other held-out family clears it. A longer list would catch both, so the detector's held-out number measures only how well its list anticipates the attacker's labels. The detector's cost is false positives: it alone flags 54.8% [50.4%, 59.1%] of the structured-benign test set (58.6% with the guard's own flags) and 87.5% of benign content carrying a forged pair. Inference-time segment-max over-flags less than the detector but more than T: it flags 9.8% of structured-benign content (T: 5.0%) and 13.0% of benign content with a benign forged pair (T: 0.7%).

Naive adversarial training. Naive adversarial training fits a LoRA on the same examples with hard 0/1 ground-truth labels. We ran it only with fixed decoys (Table 8, two seeds). There it is cleared on 0.9–1.1% of held-out-family inputs (T: 1.2–2.0%) and over-refuses none of the structured-benign set, which overlaps its hard negatives; this contradicts our pre-registered prediction (F6 in Table 7). Its AUROC is higher (0.921–0.925), likely because it learns from ground-truth labels instead of the base guard's scores, so it is a retrained detector, not a patched copy of the base guard. At $\tau = 0.5$ it flags more of everything: its clean FPR is 12.6–14.9% (T: 5.2–5.3%) and its recall is far higher (Table 8). Our result files keep no per-item clean scores, so we cannot compare the two at matched recall. We prefer T because it keeps the base guard's clean-content behaviour, which we set out to preserve, and because naive training misses trailing harm (below).

Trailing harmful content. A guard trained against rebinding could learn to stop reading at a forged pair [43], which would open a smuggling channel: benign content, a forged pair, then harmful text. The harmful-decoy column of Table 5 tests this. T and B+T flag 100% of these items with the fixed decoys and 95.5–98.7% with held-out ones. The clean-copy target flags 0.8% and naive adversarial training 8.0–8.6%. Only this test separates the clean-copy target from T.

Other guards. With the fixed decoy and query (Table 8, one seed), T transfers to Llama Guard 3 1B unchanged, cutting held-out-family bypass from 82.0% [80.4%, 83.5%] to 2.5% [1.9%, 3.2%]; on Qwen3Guard 0.6B the sampled families are mostly generic syntaxes it does not rebind to (Section 4.2), and the drop from 9.5% to 0.0% says little. The cost here is much larger than on Shieldstral. Benign content carrying a benign forged pair is flagged about ten times as often on Llama Guard 3 (2.4% to 22.5%) and more than twice as often on Qwen3Guard (10.5% to 25.9%). We do not know why.

Table 6: Adaptive attacks on Shieldstral, 50 flagged test items, seed 0. Each cell is the number of items cleared, with Wilson 95% interval. "All attacks" is the per-item worst case over the three attacks (GCG ran on 20 items and not on B+T, so that cell is a lower bound). The plain-text control (benign prompts, no forged structure) is not counted in it.

guard	score-guided search	structural search	all attacks	plain-text control
none (base)	50/50 [93%, 100%]	50/50 [93%, 100%]	50/50 [93%, 100%]	35/50 [56%, 81%]
T	47/50 [84%, 98%]	4/50 [3%, 19%]	49/50 [89%, 100%]	26/50 [39%, 65%]
B+T	34/50 [54%, 79%]	4/50 [3%, 19%]	≥34/50 (no GCG)	not run

6.3 Adaptive evaluation

The defense resists enumeration of the attack grammar but not an attacker who can see its score, and about half of that attacker's success needs no forged structure. The attacker knows the defense and sees its score (Section 2), following Carlini et al. [2]. Content cannot produce the boundary tokens, so the attacker optimises the content text, and every attack runs end to end on the defended model. We run three attacks on 50 flagged test items and report the per-item worst case (Table 6). *Score-guided search* hill-climbs on the defended guard's score within 300 guard calls per item, mutating one or two fields of the best candidate (such as syntax, position, decoy or forged query), with random restarts. Its decoys are the held-out halves of the decoy pools plus 500 benign user prompts from the development split, and its queries are every forged-query pool plus the deployment's own query; most of this material is unseen in training. *Structural search* tries all 654 forms of the full attack grammar and 30 random label mutations, with decoys and queries drawn from the held-out pools. *GCG* (Greedy Coordinate Gradient [51]) optimises a 20-token suffix inside the forged document for 250 steps, on 20 of the 50 items. A *plain-text control* runs the score-guided search with the same budget and decoy texts, but each candidate adds only the decoy text (one to five copies, before or after the request), without labels or forged query.

Both searches clear the undefended guard on every item (sanity check: at least 90%). Enumerating the grammar with held-out decoys and queries rarely beats the trained guards. It clears T and B+T on 8% [3%, 19%] of items each, within our pre-registered target of at most 20% for B+T.

The score-guided search breaks both trained guards, clearing T on 94% [84%, 98%] of items and B+T on 68% [54%, 79%]. Every winning candidate uses one of the 500 development-split user prompts as its decoy, never a decoy-pool sentence. Alone, the 35 distinct winning prompts are benign to the base guard (median score 0.0003; one, in one B+T win, scores 0.53). In 34 of T's 47 wins the best candidate places the forged structure before the request, 21 of them with an off-topic forged query. On B+T, 29 of 34 wins prepend. Almost all training attacks put the structure after the request (244 of the 252 training forms), and the search found the place training did not cover. One reading is that prepended structure attacks the grading question instead of the document. The search does not separate this from its shift to prompt-like decoys. B+T is cleared less often than T (34 against 47 of 50). This is the only result in the paper where the boundary clearly helps, and each condition has a single seed here.

The plain-text control asks how much of this is rebinding at all. Benign user prompts alone, with no labels or forged query and chosen by the same search, clear T on 26 of the 47 items the full search wins. Forged structure is needed by this search only for the other 21. The control clears the base guard on 70% [56%, 81%] of items and T on 52% [39%, 65%], always with the benign text before the request. Unlike the one fixed filler of Table 1 (0%), the control picks the best of up to 300 benign candidates per item by the guard's score. Each item's first candidate is drawn before any score is seen. It alone clears the base guard on 28% [17%, 42%] of items and T on 6% [2%, 16%]. Benign text dilutes the request a little, but most of the control's success comes from choosing among candidates with the guard's score.

GCG clears T on all 20 items it was run on, which raises the per-item worst case over all attacks against T

to 49/50 (98% [89%, 100%]). We did not run GCG on B+T, so its worst case, 34/50, is a lower bound.

7 Related Work

Structure forgery against generators and judges. The "fake completion" and "combined" attacks of Liu et al. [25] forge the end of one task and the start of another; we apply that construction to guards. In attacks descended from indirect prompt injection [15], forged template or conversation structure redirects agents and chat models [3, 8, 11, 23, 45], and injected content can manipulate the scores of LLM judges [21, 40, 41]. Those targets carry out the injected task or shift their answer. Ours grade the forged task, so success is a false "safe" verdict on content that still holds the harmful request.

Attacks on guards. Guards have been bypassed with universal perturbations [26], character-injection evasion [16] and pipeline attacks [30], and a forged judge-output block bypassed OpenAI's LLM-based detectors [29]. The closest work is concurrent: Li et al. [22] forge verdict and authority fields in a typed judge's structured input, and none of their forgeries introduces a new task to grade.

Structured-query and distillation defenses. StruQ and SecAlign [4, 5] combine reserved delimiters with training to ignore injected instructions. Meta-SecAlign [7], DefensiveTokens [6] and learnable trust-boundary delimiters [49] extend this line, and SecOPD [38] and the concurrent RAISED [12] distill the model toward its behaviour on clean context. All of them defend generative instruction-followers; we apply the idea to a classifier's content slot, and our target does not need to know which segment is the injection. Like other fine-tuning defenses [34, 36], ours falls to adaptive attacks.

8 Limitations and Conclusion

Limitations. The confirmatory study covers one guard, one English corpus and short text-only documents, and the pre-benchmark role-boundary and Llama Guard 4 checks are label-only on 30 requests (Section 4.2). Bypass counts only items the guard flags without an attack, and Shieldstral misses about half of harmful items to begin with (clean recall 47% on the test split). Bypass is an upper bound on end-to-end harm, and we did not test retrieval, tool-result or multi-turn channels. Several results rest on small or single runs: the Qwen3Guard task probe (40 items), the adaptive attacks (50 items, 20 for GCG, none on B+T), and every trained condition except T, which has two seeds. The smaller guards and naive adversarial training had fixed decoys only. On the defense side, our held-out results cover appended forms only, and the defense does nothing against searched benign prompts (Section 6.3). Segment-max keeps only the combined harm that the base guard itself detects, and we ran no composition attack. We do not claim general prompt-injection robustness [37].

Conclusion. Guards that read their task and the content as one plain-text stream can be made to grade a forged task, so a harmful request is approved unchanged. On Shieldstral and gpt-oss-safeguard the verdict follows the decoy, as rebinding predicts. Segment-max training nearly closes the enumerated appended channel with little change in clean detection, but an adaptive attacker with score access still wins, partly without rebinding. Deployers should train the guard, treat a content boundary as a cheap extra, and never let an agent act on a guard verdict alone. Defenses against rebinding should be tested against prepended structure and reported next to a plain-text control.

9 Ethics and Disclosure

Disclosure. We reported the attack to the vendors of the affected guards before writing this paper, and none had responded at the time of writing. An informal version of the attack has been public since an earlier blog post [19], so we release the paper, the benchmark and the defense without an embargo. Vendors can adopt the defense as a mitigation.

Harms and benefits. The attack is simple and transfers across guards, so we document it and offer a partial mitigation, one that nearly closes the enumerated appended channel but not adaptive attacks, instead of leaving it latent. We author no harmful text. Every harmful request, and every short example we reproduce, comes from published benchmarks (Section 4).

References

[1] Antonia Calvi, Avinash Sooriyarachchi, Giada Pistilli, Guillaume Lample, Maarten Buyl, Maximilian Augustin, Maximilian Müller, Pierre Stock, Tom Bewley, Wassim Bouaziz, et al. Shieldstral, 2026.

[2] Nicholas Carlini, Anish Athalye, Nicolas Papernot, Wieland Brendel, Jonas Rauber, Dimitris Tsipras, Ian Goodfellow, Aleksander Madry, and Alexey Kurakin. On Evaluating Adversarial Robustness, 2019.

[3] Hwan Chang, Yonghyun Jun, and Hwanhee Lee. ChatInject: Abusing Chat Templates for Prompt Injection in LLM Agents, 2025.

[4] Sizhe Chen, Julien Piet, Chawin Sitawarin, and David Wagner. StruQ: Defending Against Prompt Injection with Structured Queries, 2024.

[5] Sizhe Chen, Arman Zharmagambetov, Saeed Mahloujifar, Kamalika Chaudhuri, David Wagner, and Chuan Guo. SecAlign: Defending Against Prompt Injection with Preference Optimization, 2024.

[6] Sizhe Chen, Yizhu Wang, Nicholas Carlini, Chawin Sitawarin, and David Wagner. Defending Against Prompt Injection With a Few DefensiveTokens, 2025.

[7] Sizhe Chen, Arman Zharmagambetov, David Wagner, and Chuan Guo. Meta-SecAlign: Training LLMs against Prompt Injection for Robust Agents, 2025.

[8] Zheng Chen and Buhui Yao. Goal Hijacking Attack on Large Language Models via Pseudo-Conversation Injection, 2024. 2025 IEEE 24th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom), 2025.

[9] Jianfeng Chi, Ujjwal Karn, Hongyuan Zhan, Eric Smith, Javier Rando, Yiming Zhang, Kate Plawiak, Zacharie Delpierre Coudert, Kartikeya Upasani, and Mahesh Pasupuleti. Llama Guard 3 Vision: Safeguarding Human-AI Image Understanding Conversations, 2024.

[10] Justin Cui, Wei-Lin Chiang, Ion Stoica, and Cho-Jui Hsieh. OR-Bench: An Over-Refusal Benchmark for Large Language Models, 2024.

[11] Xinhao Deng, Jiaqing Wu, Miao Chen, Yue Xiao, Ke Xu, and Qi Li. Automating Agent Hijacking via Structural Template Injection, 2026.

[12] Mohamed Dhouib, Clement Elliker, Alexi Canesse, Maël Jenny, Lucas-Andrei Thil, Mahammed El Sharkawy, Sonia Vanier, and Elie Bursztein. RAISED: Self-Distillation for Robustness to Prompt Injection in LLM Agents, 2026.

[13] Shaona Ghosh, Prasoon Varshney, Makesh Narsimhan Sreedhar, Aishwarya Padmakumar, Traian Rebedea, Jibin Rajan Varghese, and Christopher Parisien. Aegis2.0: A Diverse AI Safety Dataset and Risks Taxonomy for Alignment of LLM Guardrails, 2025.

[14] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. The Llama 3 Herd of Models, 2024.

[15] Kai Greshake, Sahar Abdelnabi, Shailesh Mishra, Christoph Endres, Thorsten Holz, and Mario Fritz. Not what you've signed up for: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection, 2023.

[16] William Hackett, Lewis Birch, Stefan Trawicki, Neeraj Suri, and Peter Garraghan. Bypassing LLM Guardrails: An Empirical Analysis of Evasion Attacks against Prompt Injection and Jailbreak Detection Systems, 2025.

[17] Hakan Inan, Kartikeya Upasani, Jianfeng Chi, Rashi Rungta, Krithika Iyer, Yuning Mao, Michael Tontchev, Qing Hu, Brian Fuller, Davide Testuggine, et al. Llama Guard: LLM-based Input-Output Safeguard for Human-AI Conversations, 2023.

[18] Jiaming Ji, Mickel Liu, Juntao Dai, Xuehai Pan, Chi Zhang, Ce Bian, Chi Zhang, Ruiyang Sun, Yizhou Wang, and Yaodong Yang. BeaverTails: Towards Improved Safety Alignment of LLM via a Human-Preference Dataset, 2023.

[19] Utkarsh Kanwat. A safety model will grade whichever text you show it last. https://utkarshkanwat.com/writing/safety-guards-wrong-document, September 2026.

[20] Hao Li and Xiaogeng Liu. InjecGuard: Benchmarking and Mitigating Over-defense in Prompt Injection Guardrail Models, 2024.

[21] Songze Li, Chuokun Xu, Jiaying Wang, Xueluan Gong, Chen Chen, Jirui Zhang, Jun Wang, Kwok-Yan Lam, and Shouling Ji. LLMs Cannot Reliably Judge (Yet?): A Comprehensive Assessment on the Robustness of LLM-as-a-Judge, 2025.

[22] Xiang Li, Aichun Huang, Mingming Zhang, and Zheli Liu. Same Attack, Different Judge: A First Controlled Adversarial Comparison of Jev, LLM Judges, and Purpose-Built Safety Classifiers, 2026.

[23] Xitao Li, Haijun Wang, Jiang Wu, and Ting Liu. Separator Injection Attack: Uncovering Dialogue Biases in Large Language Models Caused by Role Separators, 2025.

[24] Zi Lin, Zihan Wang, Yongqi Tong, Yangkun Wang, Yuxin Guo, Yujia Wang, and Jingbo Shang. ToxicChat: Unveiling Hidden Challenges of Toxicity Detection in Real-World User-AI Conversation, 2023. EMNLP findings 2023.

[25] Yupei Liu, Yuqi Jia, Runpeng Geng, Jinyuan Jia, and Neil Zhenqiang Gong. Formalizing and Benchmarking Prompt Injection Attacks and Defenses. In *33rd USENIX Security Symposium (USENIX Security 24)*, 2024.

[26] Neal Mangaokar, Ashish Hooda, Jihye Choi, Shreyas Chandrashekaran, Kassem Fawaz, Somesh Jha, and Atul Prakash. PRP: Propagating Universal Perturbations to Attack Large Language Model Guard-Rails, 2024.

[27] Todor Markov, Chong Zhang, Sandhini Agarwal, Tyna Eloundou, Teddy Lee, Steven Adler, Angela Jiang, and Lilian Weng. A Holistic Approach to Undesired Content Detection in the Real World, 2022.

[28] Mantas Mazeika, Long Phan, Xuwang Yin, Andy Zou, Zifan Wang, Norman Mu, Elham Sakhaee, Nathaniel Li, Steven Basart, Bo Li, et al. HarmBench: A Standardized Evaluation Framework for Automated Red Teaming and Robust Refusal, 2024.

[29] Conor McCauley and Kasimir Schulz. Same Model, Different Hat. HiddenLayer research blog, https://www.hiddenlayer.com/research/same-model-different-hat, October 2025.

[30] Ian R. McKenzie, Oskar J. Hollinsworth, Tom Tseng, Xander Davies, Stephen Casper, Aaron D. Tucker, Robert Kirk, and Adam Gleave. STACK: Adversarial Attacks on LLM Safeguard Pipelines, 2025.

[31] Meta. Llama Guard 4 Model Card. https://huggingface.co/meta-llama/Llama-Guard-4-12B, April 2025.

[32] Meta. Llama Prompt Guard 2 model card. https://huggingface.co/meta-llama/Llama-Prompt-Guard-2-86M, April 2025.

[33] Mistral AI. Shieldstral. https://mistral.ai/news/shieldstral/, August 2026.

[34] Milad Nasr, Nicholas Carlini, Chawin Sitawarin, Sander V. Schulhoff, Jamie Hayes, Michael Ilie, Juliette Pluto, Shuang Song, Harsh Chaudhari, Ilia Shumailov, et al. The Attacker Moves Second: Stronger Adaptive Attacks Bypass Defenses Against Llm Jailbreaks and Prompt Injections, 2025.

[35] OpenAI. Technical Report: Research Preview of gpt-oss-safeguard. https://cdn.openai.com/pdf/08b7dee4-8bc6-4955-a219-7793fb69090c/Technical_report__Research_Preview_of_gpt_oss_safeguard.pdf, October 2025.

[36] Nishit V. Pandya, Andrey Labunets, Sicun Gao, and Earlence Fernandes. May I have your Attention? Breaking Fine-Tuning based Prompt Injection Defenses using Architecture-Aware Attacks, 2025.

[37] Dewank Pant, Shruti Lohani, and Avijit Kumar. On the Inseparability of Instructions and Data in Shared-Embedding Sequence Models, 2026.

[38] Yibo Peng, Long Lian, David Wagner, and Sizhe Chen. SecOPD: Mitigating Adaptive Prompt Injections by On-Policy Distillation, 2026.

[39] Paul Röttger, Hannah Rose Kirk, Bertie Vidgen, Giuseppe Attanasio, Federico Bianchi, and Dirk Hovy. XSTest: A Test Suite for Identifying Exaggerated Safety Behaviours in Large Language Models, 2023.

[40] Jiawen Shi, Zenghui Yuan, Yinuo Liu, Yue Huang, Pan Zhou, Lichao Sun, and Neil Zhenqiang Gong. Optimization-based Prompt Injection Attack to LLM-as-a-Judge, 2024.

[41] Manveer Singh Tamber and Jimmy Lin. Illusions of Relevance: Arbitrary Content Injection Attacks Deceive Retrievers, Rerankers, and LLM Judges, 2025.

[42] Eric Wallace, Kai Xiao, Reimar Leike, Lilian Weng, Johannes Heidecke, and Alex Beutel. The Instruction Hierarchy: Training LLMs to Prioritize Privileged Instructions, 2024.

[43] Zihao Wang, Yibo Jiang, Jiahao Yu, and Heqing Huang. The Illusion of Role Separation: Hidden Shortcuts in LLM Role Learning (and How to Fix Them), 2025.

[44] Kisu Yang, Yoonna Jang, and Heuiseok Lim. Nameless Tokenization: A Lossless Tokenizer-Level Defense Against Control-Token Forgery in Open-Weight LLMs, 2026.

[45] Charles Ye, Jasmine Cui, and Dylan Hadfield-Menell. Prompt Injection as Role Confusion, 2026.

[46] Wenjun Zeng, Yuchi Liu, Ryan Mullins, Ludovic Peran, Joe Fernandez, Hamza Harkous, Karthik Narasimhan, Drew Proud, Piyush Kumar, Bhaktipriya Radharapu, et al. ShieldGemma: Generative AI Content Moderation Based on Gemma, 2024.

[47] Yan Zhan, Yunze Song, Mengkai Hou, Wanting Zhang, Shaobo Liu, and Zhijun Gao. Same Bytes, Different Authority: Reserved-Token Representations in Chat-Template Prompt Injection, 2026.

[48] Haiquan Zhao, Chenhan Yuan, Fei Huang, Xiaomeng Hu, Yichang Zhang, An Yang, Bowen Yu, Dayiheng Liu, Jingren Zhou, Junyang Lin, et al. Qwen3Guard Technical Report, 2025.

[49] Luman Zhao, Minghui Xu, Yue Zhang, and Yijun Yang. LTBD: Learnable Trust-Boundary Delimiters for Prompt Injection Defense, 2026.

[50] Yuanbo Zhou, Changjia Zhu, Junyu Wang, Xu He, Yan Zhai, Kun Sun, Mingkui Wei, and Junjie Xiong. Prompt Overflow: What the Guardrail Inspects Is Not What the Model Infers, 2026.

[51] Andy Zou, Zifan Wang, Nicholas Carlini, Milad Nasr, J. Zico Kolter, and Matt Fredrikson. Universal and Transferable Adversarial Attacks on Aligned Language Models, 2023.

A Protocol and Pre-registered Outcomes

We froze the first protocol before the confirmatory run and did not edit it after seeing results, and we record every deviation from either protocol. These runs are exploratory and fall outside both frozen protocols: the wider set of 100 requests, the three-decoy comparison, the off-topic-query control, the forged-query (attacker knowledge) variants, the forged-verdict and split-payload controls, the real-control-token probe, the colon-label detector and label-normalisation tests, the clean-copy ablation and the GCG control without a forged pair. The second protocol was frozen before any test-split run and has four recorded deviations.

- D1 adds the variability design: three instantiations per item and family, two paraphrased deployment policies on six families, and three training seeds per defense condition.
- D2 replaces Qwen3Guard's conversation-only native forgery with the full-task forgery found on the train split, and adds the length control.
- D3 trims compute: 5,000 training items with two attacks each, three seeds for B+T and two for T and naive adversarial training, GCG on 20 items at 250 steps, and structural search on 50 items.
- D4 was recorded after a first round of test-split defense runs and before the reported ones, so it is a deviation made with test results in view. The fixed-decoy conditions of Table 8 come from that first round. D4 draws training decoys and forged queries from one half of each pool and evaluates on the other half, adds the score-guided search and the two training-free baselines, and replaces the D3 seed plan with a smaller set to save compute: two seeds for T and one for B+T and the clean-copy target.

Table 7 lists every prediction with its outcome.

Table 7: Pre-registered predictions and outcomes. P: confirmatory study; M: benchmark; F, A: defense. Intervals are 95%.

	prediction	outcome
P1	instruction-only $\leq 10\%$	confirmed (0%)
P2	forged pair $\geq 70\%$	confirmed (100%)
P3	harmful decoy lowers bypass	not a valid test as run: the panel's "harmful decoy" scores 0.03 alone, so it is the borderline decoy; bypass fell to 29–35%; the exploratory three-decoy comparison and M3 address the prediction
P4	not recency	confirmed (append 100%, prepend 15%)
P5	syntax not load-bearing	confirmed (all variants high, no labels 0%)
P6	wide policy-dependence	refuted: median 99%, range 87–100%
P7	directive does not fix	confirmed (unchanged)
P8	cross-model transfer	confirmed (30/30, 29/29)
P9	escaping fixes it	refuted: bracket-escape 88%, label-strip 91%, nonce-fence 93%
M1	a core family $\geq 70\%$ on every guard	confirmed; best family 93.4–100%
M2	instruction-only $\leq 10\%$ on every guard	not confirmed; $\leq 7.0\%$ on four guards, 20.2% on ShieldGemma
M3	dose–response, $\rho \leq -0.8$ and non-increasing bins (Shieldstral, Qwen3Guard)	partly supported on Shieldstral: $\rho = -0.84$, interval $[-0.93, -0.67]$ over decoys ($[-0.85, -0.82]$ over requests), -0.78 on the dev split, bins not monotone; not run on Qwen3Guard
M4	control-token syntax $\geq$ angle; gap closes with special parsing off	partly supported; higher in 7 of 8 comparisons but clearly so only on Shieldstral; the parsing-off arm was not part of the benchmark run
M5	exploratory: category, source, size	released with the artifacts; size not varied within a series
F1	B alone leaves held-out bypass $\geq 50\%$	confirmed; 76.8% [75.8%, 77.8%]
F2	T generalises worse than B+T on held-out families	confirmed by the pre-registered criterion, with an effect under one point; T − B+T = +0.7 points $[-0.1, +1.5]$, one-sided $p = 0.04$ ($n = 290$)
F3	B+T held-out bypass $\leq 10\%$	confirmed; 0.9% [0.7%, 1.1%]
F4	B+T AUROC within 0.01, recall within 2 points (on the interval of the difference)	not confirmed; AUROC -0.010 $[-0.041, +0.021]$, recall -0.1 $[-5.0, +4.8]$: point estimates inside the margins, intervals wider
F5	B+T over-refusal within 2 points (on the interval of the difference)	not confirmed; XSTest-safe -2.7 $[-11.1, +5.4]$, OR-Bench-hard $+0.25$ $[-6.6, +7.1]$, structured benign 0.0 $[-3.0, +3.0]$: no point estimate rises by more than 0.25, intervals wider than the margin
F6	naive adversarial training costs $\geq$ 3 points structured-benign over-refusal or generalises worse	not confirmed with the fixed decoy (0.0% on a set overlapping its hard negatives, 0.9–1.1% held-out); its costs are clean FPR and the trailing-pair blind spot
A1	GCG against B+T $\leq 30\%$	not tested on B+T; on T GCG clears 20/20
A2	structural search against B+T $\leq 20\%$	confirmed; 4/50 (8% [3%, 19%]); the score-guided search, added in D4 without a prediction, clears 34/50
A3	LLM attacker	not run

B Benchmark Grammar

The attack grammar has the axes below. The full grammar yields 659 forms (654 attacks, 5 controls), and the core panel has 63. A family is held out if its syntax is JSON, YAML, markdown bold or control token, if its language is neither English nor Spanish, or if it has at least three forged pairs. Spanish-label families are therefore held out only through the syntax or pair-count rule. The defense is trained on the remaining families.

- label syntax: guard-native tags, angle brackets, square brackets, markdown, plain colon, abbreviations, JSON, XML, YAML, control tokens;
- number of forged pairs: 1–5; position: append, prepend, middle;
- decoy harm: benign, borderline, harmful; forged query: exact, paraphrase, generic, out-of-domain; language; encoding: plain text or real control tokens.

Per-family, per-category and per-source bypass and threshold sensitivity are released with the artifacts.

C Additional Attack Analyses

Activation patching. Activation patching on Shieldstral is consistent with rebinding but does not identify the circuit that implements it. We run the attacked input and the same input with request and decoy swapped ($n = 100$), using for each request a decoy of the closest token length and matching positions by role, not by index, which keeps any remaining length mismatch from misaligning them. Patching the residual stream at the final position flips the verdict on at least 99% of inputs at every layer from 20 onward (of 26, numbered from 0). Patching the forged labels, the forged query or the decoy's positions at any layer moves the yes/no logit margin by at most 0.05 of the gap between the two inputs and flips at most 3% of verdicts.

Benchmark variations. Real control-token syntax changes bypass relative to angle-bracket labels by -1.5 to $+10.4$ points, which only partly supports our prediction that it would raise bypass. Across the three deployment policies, bypass moves by at most 6.8 points on Shieldstral, Llama Guard 3 and Qwen3Guard, and by up to 13.7 points on ShieldGemma.

Multiple verdicts from gpt-oss-safeguard. On stacked forged pairs gpt-oss-safeguard sometimes returns one verdict per content block. We count an item as flagged if any verdict is 1, the reading least favourable to the attack, chosen after the multi-verdict replies were seen. Reading only the last verdict, as a pipeline acting on the trailing label would, raises two stacked-pair families from 2.9–3.5% to 87.3–90.6% and moves every other family by under one point.

D Defense Implementation and Training

Bounded format. The bounded format renders the stock chat template with a placeholder document and splits it into a trusted prefix and suffix. The content is encoded on its own with special-token parsing disabled, and the two reserved boundary tokens (unused in pretraining) are inserted around it. The template puts a space before the document, which the tokenizer would otherwise merge into the content's first token. We move that space into the trusted prefix, so plain documents tokenize exactly as they do in the stock format (verified on 1,600 documents: AdvBench rows 200–299 and the clean test sample).

Training. We train a LoRA (rank 16, $\alpha = 32$) on the language model's attention and MLP projections with AdamW at learning rate 10^{-4}, batch 8, one epoch and maximum length 384. The loss is binary cross-entropy between the yes/no logit margin and the segment-max teacher target. LoRA does not touch the embeddings. In B+T a trainable two-row offset, initialised at zero, is added to the embeddings of the two boundary tokens (ids 36 and 37), and every other embedding stays frozen. The data are 5,000 train-split prompt items, each clean and with two training-family attacks, plus 1,250 structured-benign items, for 16,250 examples. One epoch is 2,032 steps, the last batch partial. Each attacked example's decoy and forged query are a seeded draw from the training halves of the pools: 20 of the 40 benign decoys, 3 of 6 borderline and 4 of 8 harmful decoys, and 4 of the 8 forged queries at each relatedness level. The hard negatives are train-split structured-benign items. Naive adversarial training and the clean-copy ablation use the same examples with different targets.

Compute and adaptive budgets. Training and the test-split defense, adaptive and mechanism runs used one NVIDIA RTX 4090. A training run takes about 17 minutes (1,028 s for T and 1,037 s for B+T at the last logged step, 2,030 of 2,032). The benchmark ran on an Apple-silicon laptop GPU. The score-guided search has 300 guard calls per item on 50 items, starting from 20 random candidates and then mutating batches of 10 children with restart probability 0.2. The structural search scores all 684 attacks on each of 50 items (batch 128). GCG optimises a 20-token suffix inside the forged document for 250 steps (batch 32 candidates per step) on 20 items, and stops on an item at its first success.

Fixed-decoy conditions. Table 8 gives the conditions trained and tested with one fixed decoy and query per family: naive adversarial training and the generality runs of T on Llama Guard 3 1B and Qwen3Guard 0.6B (Section 6). They have no held-out-decoy column. The hard negatives of the trained rows include 380 of the 500 structured-benign test items, so their structured-benign column is not a held-out measure.

Table 8: Fixed-decoy conditions on the test split, $\tau = 0.5$, in %. Columns as in Tables 4 and 5; naive adversarial training has two seeds (range), the other rows seed 0.

	bypass ↓		detection			over-refusal ↓			benign + forged pair	
condition	train	held-out	AUROC	recall	FPR	OR-B.	XST.	struct.	benign	harmful
naive adv. training (2 seeds)	0.9–1.0	0.9–1.1	0.921–0.925	83.3–84.6	12.6–14.9	0.3–0.8	8.0–10.7	0.0	7.8–8.2	8.0–8.6
Llama Guard 3 1B, base	84.0	82.0	0.758	58.7	16.9	40.5	1.3	23.4	2.4	97.3
Llama Guard 3 1B, T	2.3	2.5	0.771	58.4	15.0	38.3	1.3	23.0	22.5	100
Qwen3Guard 0.6B, base	28.3	9.5	0.917	88.5	23.6	76.8	21.3	30.4	10.5	100
Qwen3Guard 0.6B, T	0.0	0.0	0.924	89.0	23.3	77.3	20.0	29.4	25.9	100

OR-B.: OR-Bench-hard; XST.: XSTest-safe; struct.: structured benign; benign / harmful: decoy of the forged pair on benign content.

Content tokens. Disabling special-token parsing leaves atomic any added token that the tokenizer does not mark special [44, 47], so we checked which added-vocabulary ids the content encoder can still emit. For each guard tokenizer we encoded every added token's string on its own, embedded in plain text and in 500 random concatenations, with special-token parsing disabled, and recorded any added or special id in the output. No added or special id is reachable from content on Shieldstral, the guard the defense is trained on (1000 added tokens, all special, with the boundary tokens at ids 36 and 37), or on Llama Guard 3 1B (256, all special). On Qwen3Guard-0.6B, 12 of 26 added tokens are not marked special and stay atomic, among them the tool-call tags <tool_call> and </tool_call> and the fill-in-the-middle markers. On ShieldGemma 2B, 242 of 249 are not special (the <unused$k$> and other reserved markers) and also stay atomic. Content can produce these ids, so a boundary-token design on these tokenizers must not use them as boundary tokens.

E Reproducibility

The runs used these Hugging Face revisions: Shieldstral-1.0-3B `003ec7e`, Llama Guard 3 1B `acf7aaf`, Qwen3Guard-Gen-0.6B `fada3b2` and ShieldGemma 2B `d1dffc9`. Local guards are scored from a single forward pass with no sampling. The score is the probability of the unsafe label renormalised over the guard's label tokens, computed in fp32 from bf16 hidden states. We reached the API-served guards, gpt-oss-safeguard-20B and Llama Guard 4 12B, through OpenRouter at temperature 0 without pinning a provider. Every benchmark call to gpt-oss-safeguard was served by Groq, as recorded in the response metadata; the cross-model runs did not record the provider. Dataset sources and licenses: AdvBench, HarmBench, Aegis 2.0, BeaverTails, the OpenAI moderation evaluation and ToxicChat, with XSTest and OR-Bench as over-refusal sources. We use the non-commercial sets (ToxicChat, BeaverTails) for research only. All hyperparameters, splits and the full protocol are in Sections A and D.

F Open Science

We release the GuardRebind benchmark and harness, the attack grammar, the defense code, the training commands and data recipe, and the result files behind every number in this paper, at `https://github.com/ukanwat/guardrebind`. The released results contain item ids and scores, plus text only from the public benchmarks we used (such as the AdvBench requests of the confirmatory run and the benign development-split prompts that won as search decoys) and model outputs that quote it. We release no new harmful content. The pre-registered protocol and its deviations are included so that predictions can be checked against outcomes.